

OOP Project Proposal

BS Data Science

1. Project Title

AeroNexus – Flight Management System

2. Team Members

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3. Project Idea & OOP Concepts

Problem / Idea Description

A Flight Management System that allows an airline company to manage its fleet and flight schedules through a console or GUI-based interface. The system addresses the real-world need for organized record-keeping of aircraft and flight data — enabling staff to use CRUD operations on planes and flights databases, with all data persisted across sessions via CSV files.

OOP Concepts Used

OOP Concept	How it appears in your project
Classes & Objects	Company, DataBase, PlaneDB and FlightDB
Encapsulation	All data members of each class are Private.
Inheritance	PlaneDB and Flight DB are child classes of Database class.
Polymorphism	PlaneDB & FlightDB must inherit and override functions like Load(), Update(), Record(), Display(), etc.
Constructors	Caompany constructor initializes the data bases. DB constructors initialize dynamic arrays to store complete data present in the CSV file.

Why This Project Is Useful

Airlines and aviation companies rely on accurate, up-to-date records of their fleet and flight schedules. This system replaces error-prone manual tracking with a structured, persistent solution that keeps plane and flight data organized and easily retrievable.

4. Tools & Technologies

Language: C++ | **IDE:** VS Code | **Interface:** Console-based menu-driven / GUI-based button operated | **Data Storage:** File handling via two CSV files (planes.csv, flights.csv)